

Lecture 7

A Brief Review of Generalized Linear Models

7.1

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See Fahrmeir L., Kneib T., Lang S. (2007) Regression. Modelle, Methoden und Anwendungen, Springer. (2013) Regression. Models, Methods and Applications, Springer.

7.2

Classical linear regression models cannot be used when the outcome is not a metric variable.

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1 Binary Regression Models

Data

$$(y_i, x_{i1}, \dots, x_{ip}), \quad i = 1, \dots, n,$$

with y binary (0/1) variable, obtained by recoding a dichotomous categorical outcome, and $x_1, \dots, x_p$ metric predictors or categorical regressors recoded as binary variables.

The outcomes

$$(y_i | x_{i1}, \dots, x_{ip}) \text{ and } (y_j | x_{j1}, \dots, x_{jp}), \quad i \neq j$$

are assumed to be independent.

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Model

The probability $\pi_i = P(y_i = 1 | x_{i1}, \dots, x_{ip})$ and the *linear predictor*

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = \mathbf{x}_i' \boldsymbol{\beta}, \quad i = 1, \dots, n,$$

are linked by the so-called inverse-link function $h(\eta) \in [0, 1]$:

$$\pi_i = h(\eta_i).$$

Logit Model:

$$\pi = \frac{\exp(\eta)}{1 + \exp(\eta)} \quad \leftrightarrow \quad \log\left(\frac{\pi}{1 - \pi}\right) = \eta.$$

Probit Model:

$$\pi = \Phi(\eta) \quad \leftrightarrow \quad \Phi^{-1}(\pi) = \eta.$$

Complementary log-log Model:

$$\pi = 1 - \exp(-\exp(\eta)) \quad \leftrightarrow \quad \log(-\log(1 - \pi)) = \eta.$$

Interpretation of parameters in a logit model

Correspondingly to the linear predictor

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = \mathbf{x}_i' \boldsymbol{\beta}, \quad i = 1, \dots, n,$$

the odds

$$\frac{\pi_i}{1 - \pi_i} = \frac{P(y_i = 1 | \mathbf{x}_i)}{P(y_i = 0 | \mathbf{x}_i)}$$

can be modelled according to the multiplicative relationship

$$\frac{P(y_i = 1 | \mathbf{x}_i)}{P(y_i = 0 | \mathbf{x}_i)} = \exp(\beta_0) \cdot \exp(\beta_1 x_{i1}) \cdot \dots \cdot \exp(\beta_p x_{ip}).$$

For example, if x_{i1} changes to $(x_{i1} + 1)$ the following relationship holds (ceteris paribus) for the odds ratio

$$\frac{\frac{P(y_i=1|x_{i1}+1,\dots)}{P(y_i=0|x_{i1}+1,\dots)}}{\frac{P(y_i=1|x_{i1},\dots)}{P(y_i=0|x_{i1},\dots)}} = \exp(\beta_1).$$

Therefore

- if $\beta_1 > 0$ the odds $\frac{P(y_i=1)}{P(y_i=0)}$ increases,
- if $\beta_1 < 0$ the odds $\frac{P(y_i=1)}{P(y_i=0)}$ decreases,
- if $\beta_1 = 0$ the odds $\frac{P(y_i=1)}{P(y_i=0)}$ does not change.

2 Example. Credit data

- Y: a factor with levels `buen` and `mal`, the response variable. `buen` is the good payers.
- Cuenta: a factor with levels `no`, `good running` and `bad running`, quality of the credit clients bank account.
- Mes: a numeric vector, duration of loan in months.
- Ppag: a factor with levels `pre buen pagador` and `pre mal pagador`, if the client previously have been a good or bad payer.
- Uso: a factor with levels `privado` and `profesional`, the use to which the loan is made.
- DM: a numeric vector, the size of loan in german marks.
- Sexo: a factor with levels `mujer` and `hombre`, sex of the client.
- Estc: a factor with levels `no vive solo` and `vive solo`, civil state of the client.

Source: Fahrmeir and Tutz, Multivariate Generalized Linear Models, Springer Verlag.
Data reading

```
library(Fahrmeir)
data(credit)
```

Data inspection

```
summary(credit)
```

```
##      Y      Cuenta      Mes      Ppag
##  buen:700    no      :274    Min.   : 4.0    pre buen pagador:911
##  mal :300    good running:394    1st Qu.:12.0    pre mal pagador : 89
##                bad running :332    Median :18.0
##                Mean      :20.9
##                3rd Qu.:24.0
##                Max.     :72.0
##      Uso      DM      Sexo      Estc
##  privado :657    Min.   : 250    mujer :402    no vive solo:640
##  profesional:343    1st Qu.: 1366    hombre:598    vive solo :360
##                Median : 2320
##                Mean    : 3271
##                3rd Qu.: 3972
##                Max.    :18424
```

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Observe that the baseline level (reference category) for Y is buen; so in the generalized modelling regression framework we have

$$y = \begin{cases} 0 & \text{buen} \\ 1 & \text{mal} \end{cases}$$

The reference category for Cuenta is no. In order to set bad running as reference category (see Fahrmeir et al.), it is sufficient to redefine the factor by using bad running as first declared level.

```
credit$Cuenta <- factor(credit$Cuenta, levels = c("bad running",
  "no", "good running"))
```

The size of the loan DM is transformed in thousands of Deutsche Mark, in order to reproduce the results by Fahrmeir.

```
credit$DM <- credit$DM/1000
```

The function transform can also be used to perform variable transformations.

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The first specification models the logit

$$\log \left(\frac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} \right)$$

as linear function of variables Cuenta, Mes, DM, Ppag and Uso.

```
a <- glm(Y ~ Cuenta + Mes + DM + Ppag + Uso,
  family = binomial(link = "logit"), data = credit)
```

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```
summary(a)
##
## Call:
## glm(formula = Y ~ Cuenta + Mes + DM + Ppag + Uso, family = binomial(link
##      data = credit)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8876  -0.8440  -0.4628   0.9629   2.3620
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.696004    0.197855  -8.572 < 2e-16 ***
## Cuentano         0.617659    0.174728   3.535 0.000408 ***
```

```
## Cuentagood running -1.337748 0.201127 -6.651 2.91e-11 ***
## Mes 0.033233 0.007746 4.290 1.78e-05 ***
## DM 0.023455 0.032770 0.716 0.474134
## Ppagpre mal pagador 0.986066 0.250891 3.930 8.49e-05 ***
## Usoprofesional 0.425536 0.158272 2.689 0.007174 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1221.7 on 999 degrees of freedom
## Residual deviance: 1029.8 on 993 degrees of freedom
## AIC: 1043.8
##
## Number of Fisher Scoring iterations: 4
```

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The linear effect of the size of the loan does not result statistically significant.
In the second specification of the logit model we add the square of loan duration, Mes, and size, DM.

```
b <- glm(Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) + Ppag +
  Uso, family = binomial(link = "logit"), data = credit)
```

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```
summary(b)
##
## Call:
## glm(formula = Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) + Ppag +
##      Uso, family = binomial(link = "logit"), data = credit)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4157  -0.8124  -0.4719   0.9585   2.6326
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.8870929  0.3200503  -5.896 3.72e-09 ***
## Cuentano        0.6178347  0.1761733   3.507 0.000453 ***
## Cuentagood running -1.3374022  0.2024364  -6.607 3.93e-11 ***
## Mes            0.0921909  0.0252941   3.645 0.000268 ***
## I(Mes^2)       -0.0009094  0.0004133  -2.200 0.027781 *
## DM            -0.2641266  0.0985212  -2.681 0.007342 **
## I(DM^2)        0.0229695  0.0074761   3.072 0.002124 **
## Ppagpre mal pagador 0.9953315  0.2551618   3.901 9.59e-05 ***
## Usoprofesional  0.4039789  0.1601534   2.522 0.011654 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1221.7 on 999 degrees of freedom
## Residual deviance: 1017.4 on 991 degrees of freedom
## AIC: 1035.4
##
```

```
## Number of Fisher Scoring iterations: 4
```

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The two model specifications can be compared by performing a likelihood ratio test or a Wald test.

Both tests do not reject the whole contribution of linear and quadratic terms.

```
anova(a, b, test = "LRT")
## Analysis of Deviance Table
##
## Model 1: Y ~ Cuenta + Mes + DM + Ppag + Uso
## Model 2: Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) + Ppag + Uso
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         993      1029.8
## 2         991      1017.4  2    12.444 0.001985 **
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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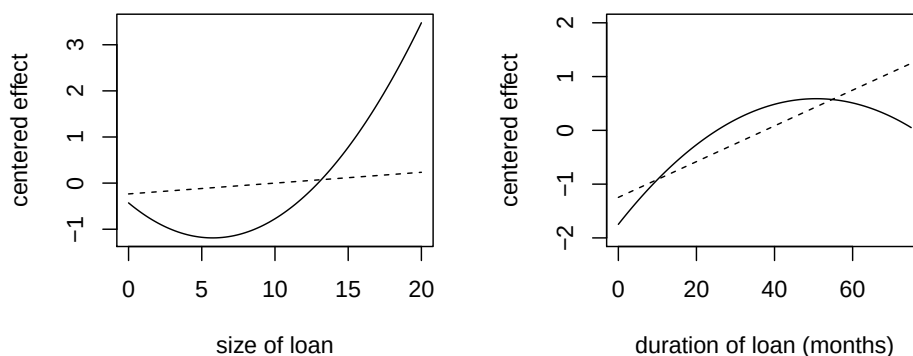
```
library(lmtest)
waldtest(a, b, test = "Chisq")
## Wald test
##
## Model 1: Y ~ Cuenta + Mes + DM + Ppag + Uso
## Model 2: Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) + Ppag + Uso
##   Res.Df Df   Chisq Pr(>Chisq)
## 1      993
## 2      991  2 11.474   0.003225 **
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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The following figures show the estimated linear and non-linear effects of the size and the duration of the loan.

It is interesting to note how the logit of being a bad payer increases in presence of loans of very small size or of big size.

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Graphs can be obtained with the following code.

```
f <- function(x) coef(b)[6] * x + coef(b)[7] * x^2
fl <- function(x) coef(a)[5] * x
fmean <- mean(f(seq(0, 20, length = 200)))
flmean <- mean(fl(seq(0, 20, length = 200)))
fc <- function(x) -fmean + coef(b)[6] * x + coef(b)[7] * x^2
flc <- function(x) -flmean + coef(a)[5] * x

g <- function(x) coef(b)[4] * x + coef(b)[5] * x^2
gl <- function(x) coef(a)[4] * x
gmean <- mean(g(seq(0, 75, length = 200)))
glmean <- mean(gl(seq(0, 75, length = 200)))
gc <- function(x) -gmean + coef(b)[4] * x + coef(b)[5] * x^2
glc <- function(x) -glmean + coef(a)[4] * x

layout(matrix(1:2, 1, 2))
curve(fc, xlim = c(0, 20), xlab = "size of loan", ylab = "centered effect")
curve(flc, xlim = c(0, 20), add = TRUE, lty = 2)
curve(gc, xlim = c(0, 75), ylim = c(-2, 2), xlab = "duration of loan (months)", ylab = "centered effect")
curve(glc, xlim = c(0, 75), add = TRUE, lty = 2)
```

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Marginal effects at the mean that describe the expected absolute change of $P(y = 1|x)$ corresponding to a unitary increase of a single predictor (ceteris paribus) can be obtained with the function `logitmfx` available in the package `mfx`.

```
library(mfx)
logitmfx(Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) + Ppag +
  Uso, data = credit, atmean = TRUE, robust = TRUE)
```

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```
library(mfx)
logitmfx(Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) + Ppag + Uso, data = credit,
  atmean = TRUE, robust = TRUE)

## Call:
## logitmfx(formula = Y ~ Cuenta + Mes + I(Mes^2) + DM + I(DM^2) +
##      Ppag + Uso, data = credit, atmean = TRUE, robust = TRUE)
##
## Marginal Effects:
##
##              dF/dx   Std. Err.      z    P>|z|
## Cuentano          1.2567e-01  3.7449e-02  3.3556 0.0007918 ***
## Cuentagood running -2.3685e-01  3.1267e-02 -7.5752 3.586e-14 ***
## Mes               1.7657e-02  4.8584e-03  3.6344 0.0002787 ***
## I(Mes^2)          -1.7418e-04  8.0922e-05 -2.1525 0.0313596 *
## DM                -5.0588e-02  2.0993e-02 -2.4097 0.0159660 *
## I(DM^2)           4.3993e-03  1.7283e-03  2.5454 0.0109159 *
## Ppagpre mal pagador 2.2132e-01  6.6821e-02  3.3121 0.0009261 ***
## Usoprofesional     7.9621e-02  3.2468e-02  2.4523 0.0141938 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## dF/dx is for discrete change for the following variables:
##
## [1] "Cuentano"           "Cuentagood running"  "Ppagpre mal pagador"
## [4] "Usoprofesional"
```

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